Charilaos Mylonas, Ph.D.

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About

I am a data science and scientific computing expert, with a strong mathematical and high performance computing background, and more than 7 years of machine learning / deep learning experience. I hold a Ph.D. on Machine Learning for Structural Health Monitoring, with original contributions on the use of deep learning and generative ML in predictive maintenance. I have extensive demonstrable experience in setting up and executing large scale Monte-Carlo simulations, working with industrial monitoring data (Wind turbine SCADA data).

Work Experience

Sept 2024 – Jan 2025 **Deloitte** Assistant Manager

- Implemented a machine learning-enhanced methodology for improving the effectiveness of compliance monitoring.

Feb 2022 – Sept 2024 **Deloitte** Senior Consultant

- Designed and created GenAI prototypes with retrieval augmented generation.

Sept 2016–Nov 2021 **ETH Zurich** Ph.D. Candidate/Research Assistant

- Researched scalable probabilistic machine learning for structural condition monitoring of wind turbines and wind farms (Python, TensorFlow).

Dec 2015–Sept 2016 **ETH Zurich** Research Assistant

- Contributed to the popular computational statistics software UQLab by implementing uncertainty quantification and sensitivity analysis algorithms

Jul 2014–Dec 2014 **Credit Suisse** Full-Stack Trading Tool Developer (internship)

- Implemented and validated a high level interface for an option pricer (C++, R), achieving more than 10-fold improvement by replacing pre-existing interface.

Education

Sept 2016 – Sept 2021 **ETH Zurich**
Ph.D. in Machine Learning for Structural Health Monitoring under Uncertainty
Advisor: Prof. Eleni Chatzi

Sept 2012 – Sept 2015 **ETH Zurich**
M.Sc. in Computational Science and Engineering
Specialization: Computational Electromagnetics
Advisor: Prof. Ralf Hiptmair

Technical Strengths

Programming Languages	Python, Matlab, R	●●●●●●
	C++, Java, JavaScript	●●●●○○
Other software development skills	Linux, Docker, Kubernetes, Classical ML Algorithms, Scientific Computing, Software Design, Web Development, High Performance Computing, Retrieval Augmented Generation systems, Microcontroller Programming	
Deep Learning	Probabilistic Generative Models (GANs, VAEs, Normalizing Flows, Denoising Diffusion models), Graph Neural Networks, Strong familiarity of all core Deep Learning architectures (gated RNNs, CNNs, Attention Mechanisms & Transformers) and how they apply to different data modalities (text, audio, images, tabular data).	

Other Information

Teaching assistant roles

· High Performance Computing for CSE (C++, OpenMP) (2020) (Prof. O. Schenk).
· Method of Finite Elements (Matlab, Python) (2017 – 2019) (Prof. E. Chatzi).

Other academic engagements

· *Mentorship*: Serving as mentor for Ph.D. students at ETH Zurich (upon invitation).
· *Student project supervision*: 6 MSc theses and semester projects and consulted on several others.
· *Reviewer assignments*: for Mechanical Systems and Signal Processing and Journal of Sound and Vibration.

Distinctions and certificates

· **Best paper award** in 39th IMAC conference (Feb. 2021).
· **SIAM Gene Golub Scholarship** for Ph.D. summer school on "*High-Performance Data Analytics*" *Aussois, France 2019.*

Selected Publications

Please refer to Google Scholar [link] for full list and updated citation count.

Mylonas, C. *(ETH Ph.D. Dissertation)* Machine Learning for Structural Health Assessment under Uncertainty, with applications in Wind Energy, [link]

Mylonas C*, Chatzi E.* Remaining Useful Life Estimation for Engineered Systems Operating under Uncertainty with Causal GraphNets. Sensors. 2021; 21(19):6325. https://doi.org/10.3390/s21196325

Mylonas, C.*, Abdallah, I., Chatzi, E.* Conditional variational autoencoders for probabilistic wind turbine blade fatigue estimation using SCADA data. Wind Energy. 2021; 1- 18. https://doi.org/10.1002/we.2621

Lai, Z., ***Mylonas, C.****, Nagarajaiah, S., & Chatzi, E.* Structural identification with physics-informed neural ordinary differential equations. Journal of Sound and Vibration, 508, 116196.

Mylonas, C.*, Abdallah, I., Chatzi, E.* (2021) Relational VAE: A Continuous Latent Variable Model for Graph Structured Data [link]

Mylonas, C.*, Tsialiamanis, G., Worden, K. & Chatzi, E.* Bayesian graph neural networks for strain-based crack localization. (*39th IMAC conference proc.*) [link]

Mylonas, C.*, Abdallah, I., & Chatzi, E. (2020).* Deep Unsupervised Learning For Condition Monitoring and Prediction of High Dimensional Data with Application on Windfarm SCADA Data. *In Model Validation and Uncertainty Quantification, Volume 3 (pp. 189-196). Springer, Cham.*